

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Industry Problem Solving Task

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA1558

COURSE OUTCOME COVERED

CO4: *Analyze big data characteristics and challenges across domains including security and application aspects.*
(BL4)

Dave's Taxonomy: Articulation (Level 4)
Question Count: 5

Total Marks: 40

Duration :60 Minutes

Code of Conduct

I AB. Dhaya shree (192472002) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

Assessment Tasks

Industry Problem Solving Task

- Retail Data Optimization Problem** - A retail chain operates both physical stores and an online platform. It generates large volumes of sales, inventory, and customer behavior data. The company wants real-time inventory tracking and demand forecasting. It also aims to deliver personalized marketing recommendations to customers. Design a big data architecture to support these requirements. Explain the tools, technologies, and analytics models you would use.
- Social Media Fraud Detection Problem** - A social media platform processes billions of user interactions daily. These include posts, comments, likes, and shares. The platform wants to detect fake accounts and misinformation quickly. Real-time or near real-time processing is required. Design a scalable big data solution for this use case. Explain the role of machine learning and stream processing frameworks.
- Government Data Platform Problem** A government agency is building a national data platform. It combines census, economic, and geospatial datasets. Different departments need access to shared and integrated data. The system must ensure interoperability across multiple agencies. Strict privacy and security requirements must be followed. Propose a governance and data management strategy.
- Banking Fraud Detection Problem**
A bank processes millions of financial transactions every minute. It needs to detect fraudulent activities in real time. The system must analyze historical and streaming transaction data. Low latency and high accuracy are critical requirements. Design a big data pipeline for fraud detection. Explain the technologies and algorithms you would implement.

5. Agriculture Smart Farming Problem

A smart farming system collects soil, weather, and crop data. Sensors and drones provide real-time field information. Farmers want insights to improve yield and resource usage. Data is generated from multiple distributed sources. Design a big data solution for precision agriculture. Explain how analytics can support decision-making.

Industry Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Industry Context	20%	Demonstrates strong understanding of real-world problem context	Good understanding	Basic understanding	Weak understanding
Application of Big Data Concepts	25%	Applies highly relevant big data ideas and tools	Mostly appropriate application	Partial application	Weak application
Solution Design	25%	Solution is practical, scalable, and well-reasoned	Reasonable solution	Basic solution with gaps	Unclear solution
Security / Risk Awareness	15%	Strong awareness of security and operational risks	Reasonable awareness	Limited awareness	Poor awareness
Clarity and Professionalism	15%	Well-structured and professional response	Generally clear	Moderately clear	Poorly presented

Total Marks Awarded:

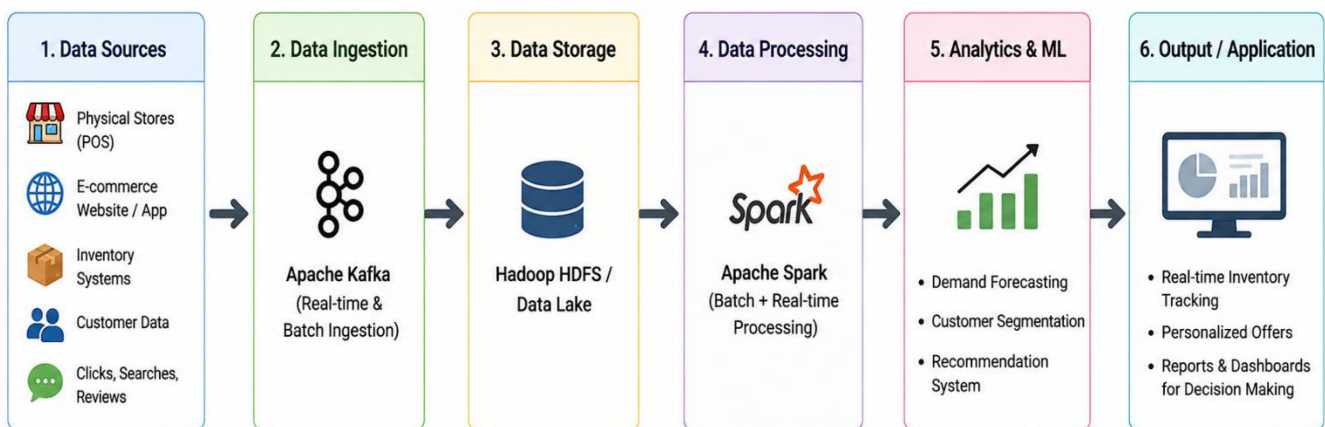
1. Retail Data Optimization Problem

Problem Understanding

A retail company operates both physical stores and an online shopping platform. It generates a huge amount of data from sales, inventory, customer searches, purchases, reviews, and website activity. The company needs real-time inventory tracking, demand forecasting, and personalized recommendations.

Proposed Big Data Architecture

Retail Data Optimization – Big Data Flow



1.Data Sources Data is collected from:

- Physical store billing/POS systems
- Online shopping website and mobile application
- Inventory and warehouse systems
- Customer transactions
- Customer clicks, searches, and reviews
- Product and supplier information

2. Data Ingestion Tools such as Apache Kafka can collect continuous streams of sales and customer activity data. Batch data can also be loaded periodically using tools such as Apache Sqoop or cloud-based data ingestion services.

3. Data Storage A distributed storage system such as Hadoop HDFS or a cloud data lake can store large amounts of structured and unstructured data.

For example:

Sales data → structured storage

Customer reviews → text data

Website clicks → log data

Product images → unstructured data

4. Data Processing Apache Spark can process both historical and real-time data quickly. Spark Streaming can process live sales and inventory events.

For example, when a product is sold, the inventory count can be updated immediately.

5. Analytics and Machine Learning Different analytics models can be used:

Demand Forecasting: Machine learning models such as Random Forest, XGBoost, or time-series models can predict future product demand.

Customer Recommendation: Collaborative filtering and recommendation algorithms can identify products that a customer may like based on previous purchases and browsing behavior.

Customer Segmentation: Clustering algorithms such as K-Means can divide customers into groups based on their shopping behavior.

6. Output The results can be displayed through dashboards using tools such as Power BI or Tableau.

- Managers can see:
- Current stock
- Fast-selling products
- Expected future demand
- Customer preferences
- Products that need restocking
- Security Customer information should be protected using encryption, access control, authentication, and data masking.

Conclusion This architecture helps the retailer track inventory in real time, predict demand, reduce stock shortages, and provide personalized recommendations. Big Data and machine learning together make the retail system faster and more intelligent.

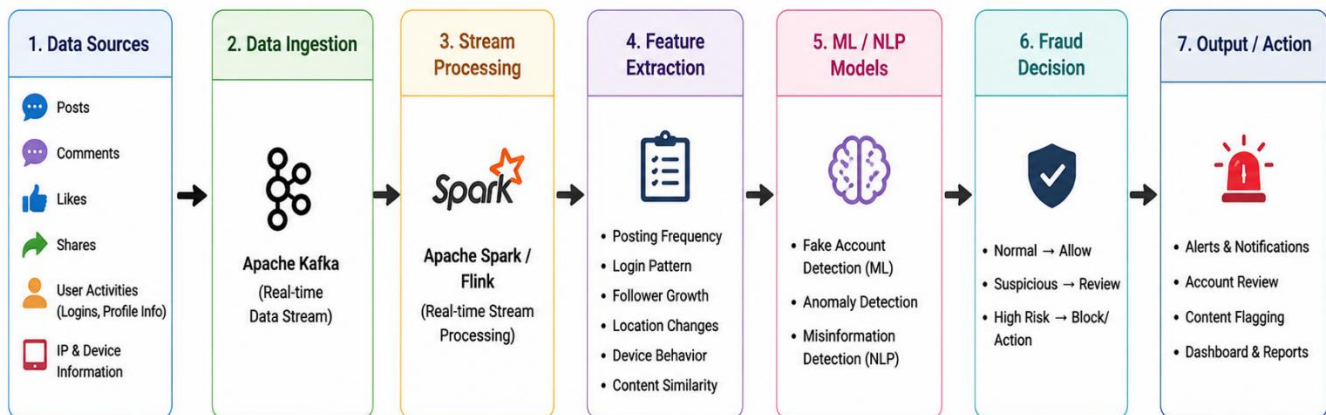
2.Social Media Fraud Detection Problem

Problem Understanding

A social media platform processes billions of posts, comments, likes, shares, and user activities every day. Fake accounts and misinformation must be detected quickly. Since the data is continuously generated, the system requires real-time or near-real-time processing.

Proposed Big Data Architecture

Social Media Fraud Detection – Big Data Architecture



1. Data Sources The system collects:

- Posts
- Comments
- Likes
- Shares
- Login information
- IP addresses
- Device information
- Followers and following relationships
- User activity patterns

2. Stream Data Ingestion Apache Kafka can receive billions of user events continuously.

For example, when a user creates a post or shares information, the event is immediately sent to Kafka.

3. Stream Processing Apache Spark Streaming or Apache Flink can process the incoming events in real time.

- The system can calculate features such as:
- Number of posts per minute
- Number of accounts contacted
- Sharing frequency
- Login location changes
- Number of followers gained quickly

4. Machine Learning for Fake Account Detection Machine learning can identify abnormal user behavior.

- Possible algorithms include:

- Random Forest
- XGBoost
- Logistic Regression
- Neural Networks
- Isolation Forest for anomaly detection

For example, an account that creates hundreds of posts within a few minutes and sends requests to thousands of users can receive a high fraud score.

5. Misinformation Detection Natural Language Processing can analyze the text of posts.

- NLP models can identify:
- Suspicious keywords
- Repeated misleading content
- Similar fake posts
- Toxic or harmful content
- Transformer-based models such as BERT can be used for text classification.

6. Real-Time Decision The system can assign a risk score to each activity.

For example:

Risk Score > 90% → Block/Review

Risk Score 60–90% → Send for verification

Risk Score < 60% → Normal activity

Security The platform should protect user information using encryption, authentication, access control, and secure APIs.

Conclusion Kafka provides scalable event collection, while Spark or Flink provides real-time processing. Machine learning and NLP identify suspicious accounts and misinformation quickly. This makes the social media platform safer and more reliable.

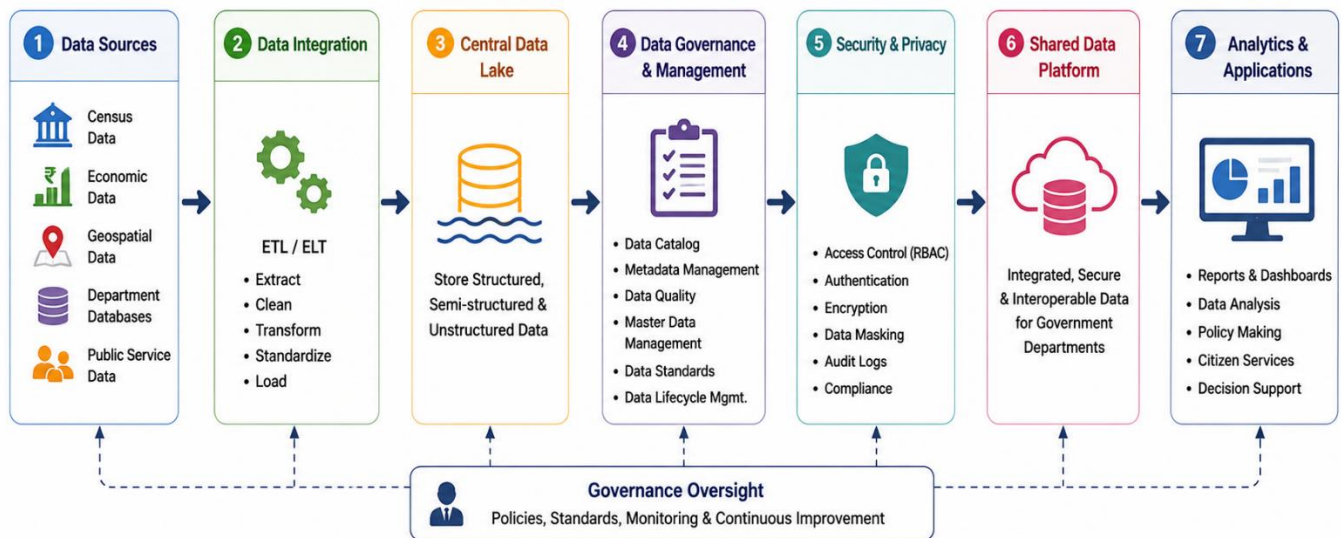
3. Government Data Platform Problem

Problem Understanding

A government agency needs to combine census, economic, and geographical data from different departments. Since multiple agencies use different systems, the main challenges are data integration, interoperability, privacy, security, and governance.

Proposed Architecture

Government Data Platform – Big Data Flow



1. Data Sources The platform can collect:

Census data

Economic data

Tax information

Population information

Geographic/GIS data

Public service data

Administrative records

2. Data Integration Different departments may use different formats and databases.

An ETL/ELT system can:

Extract data

Clean data

Transform data

Standardize data

Load it into the central platform

Common formats such as CSV, JSON, XML, and standardized APIs can improve interoperability.

3. Central Data Lake A government data lake can store large amounts of structured and unstructured information.

Historical data can be stored in distributed storage, while frequently used information can be stored in databases or data warehouses.

4. Data Governance A strong governance framework should define:

Who owns each dataset

Who can access it

How data should be used

Data quality standards

Data retention rules

Data sharing rules

A data catalog can help departments discover available datasets.

5. Data Quality Data should be checked for:

Missing values

Duplicate records

Incorrect formats

Inconsistent names

Invalid geographic information

Master Data Management can ensure that the same entity is represented consistently across departments.

6. Security and Privacy Because government data may contain sensitive information, strong security is essential.

The system should use:

Role-Based Access Control

Multi-factor authentication

Encryption

Data masking

Audit logs

Secure APIs

Network security

Regular security monitoring

Only authorized employees should access restricted datasets.

7. Interoperability Different departments should communicate using standardized APIs and common data formats.

For example, a health department and economic department should be able to exchange approved data without changing their complete internal systems.

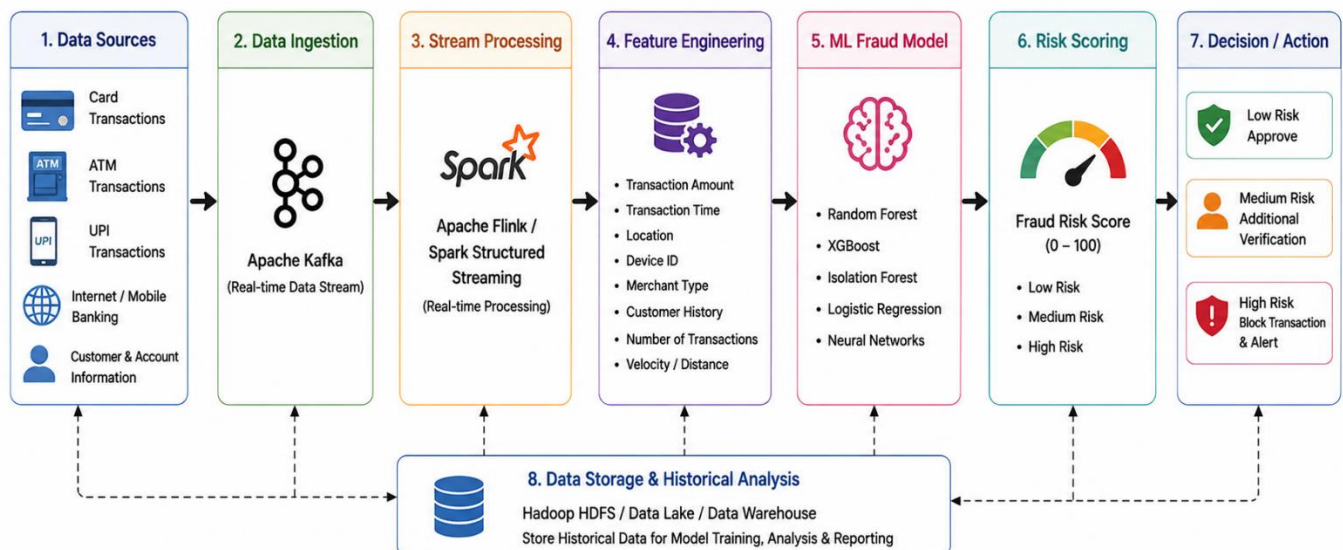
Conclusion A centralized but well-governed Big Data platform allows government departments to securely share and analyze information. Strong governance, standardization, privacy, and access control ensure that data can be used effectively without compromising citizen privacy.

4. Banking Fraud Detection Problem

Problem Understanding

A bank processes millions of transactions every minute. Fraud must be detected almost immediately because even a few seconds of delay can result in financial loss. The system must analyze both historical and real-time transaction data.

Banking Fraud Detection – Big Data Architecture



1.Data Sources The system receives:

- Credit/debit card transactions
- ATM transactions
- UPI transactions

- Online banking transactions
- Customer account information
- Device information
- IP address
- Location
- Historical transaction records

2. Real-Time Data Ingestion Apache Kafka can collect millions of transactions continuously.

Kafka is useful because it can handle high-volume data and distribute it across multiple processing systems.

3. Stream Processing Apache Flink or Spark Structured Streaming can analyze transactions immediately.

For example, if a customer normally spends ₹2,000 in India but suddenly a transaction of ₹2,00,000 occurs from another country, the system can identify it as suspicious.

4. Feature Engineering Important features can include:

- Transaction amount
- Transaction time
- Location
- Device ID
- Number of transactions
- Previous transaction behavior
- Distance between transactions
- Merchant type
- Failed transaction attempts
- Machine Learning Algorithms Random Forest / XGBoost: Useful for classification of normal and fraudulent transactions.
- Logistic Regression: Provides a simple and interpretable fraud probability.
- Isolation Forest: Useful for identifying unusual transactions.
- Neural Networks: Can learn complex patterns from large transaction datasets.

6. Real-Time Risk Score Each transaction receives a fraud probability.

For example:

Low Risk → Approve

Medium Risk → Ask for additional verification

High Risk → Block transaction and alert bank

7. Historical Data Historical transactions can be stored in a data lake or data warehouse. This data is used to train and periodically update machine learning models.
8. Low-Latency Requirement To achieve low latency:

- Kafka handles high-speed event ingestion.
- Flink/Spark processes streaming data.
- In-memory databases can store frequently required information.
- ML models can be deployed close to the transaction system.
- Security Banking data must use encryption, strong authentication, access control, audit logging, and secure communication.

Conclusion

The combination of Kafka, stream processing, Big Data storage, and machine learning provides a scalable fraud detection system. It can analyze transactions in real time and stop suspicious transactions before significant financial damage occurs.

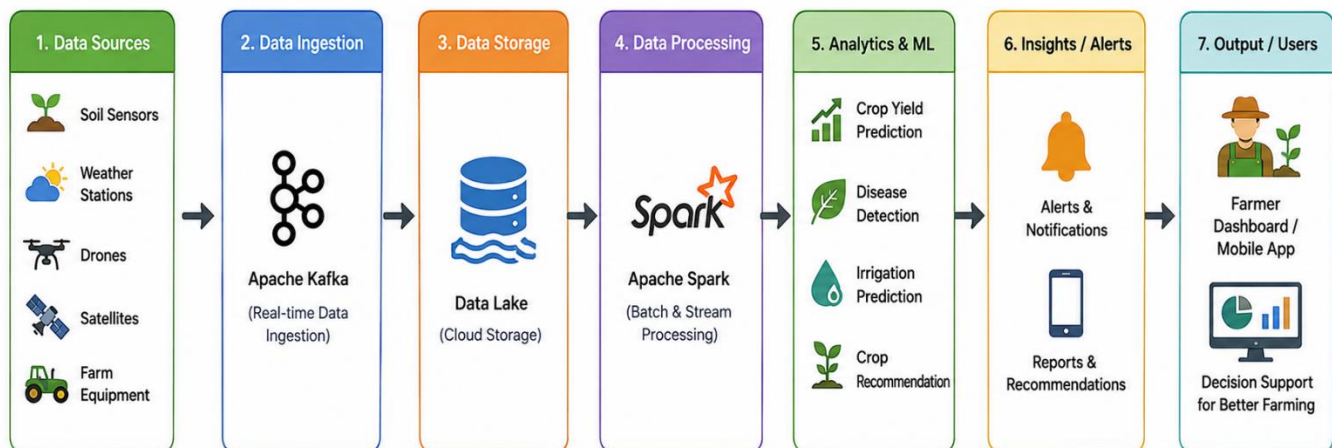
5.Agriculture Smart Farming Problem

Problem Understanding

Smart farming collects data from soil sensors, weather stations, drones, satellites, and agricultural equipment. Since these sources are distributed and continuously generate data, Big Data technology can help farmers make better decisions.

Proposed Architecture

Agriculture Smart Farming – Big Data Flow



1. Data Sources The system collects:

- Soil moisture
- Soil temperature
- Soil nutrients
- Weather information
- Rainfall

- Humidity
- Crop images
- Drone images
- Satellite images
- Irrigation information
- Crop growth information

2. Data Collection IoT sensors continuously send field information to an IoT gateway.

For example, a soil-moisture sensor can send information every few minutes.

3. Data Ingestion Apache Kafka can collect sensor and other real-time data.

This allows large numbers of sensors to send information simultaneously.

4. Data Storage A cloud data lake or Hadoop-based storage system can store:

- Sensor data
- Historical crop data
- Drone images
- Satellite images
- Weather data

5. Data Processing Apache Spark can process large amounts of agricultural data.

It can combine soil, weather, crop, and historical yield information to find useful patterns.

6. Machine Learning and Analytics

- Crop Yield Prediction: Machine learning can predict expected crop yield using soil, weather, and historical data.
- Irrigation Prediction: The system can determine when irrigation is required based on soil moisture, temperature, and weather forecasts.
- Disease Detection: Computer vision models can analyze drone or crop images and identify signs of disease.
- Crop Recommendation: Machine learning can recommend suitable crops based on soil and environmental conditions.

7. Decision Support The farmer can receive recommendations through a mobile application or dashboard.

For example:

- Soil moisture is low → Start irrigation
- Heavy rain predicted → Delay irrigation
- Disease detected → Inspect affected area
- Low soil nutrient level → Apply required fertilizer

8. Benefits The system can help:

- Increase crop yield
- Reduce water usage
- Reduce fertilizer waste
- Detect diseases early
- Reduce farming costs
- Improve resource utilization
- Security Sensor and farm data should be protected using authentication, encryption, secure communication, and access control.

Conclusion

Big Data and IoT allow farmers to understand field conditions in real time. Machine learning converts this data into useful recommendations, helping farmers improve crop yield while reducing water, fertilizer, and other resource usage.